Improving Probabilistic Diffusion Models With Optimal Diagonal Covariance Matching

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01 Motivation And Problem Formulation

Why Improve Diffusion Covariance?

Heuristic or fixed covariances require many steps for high-quality samples and likelihoods

Misspecified covariance hurts small-step regimes: worsened FID, recall, and ELBO-based likelihood

Goal: accurate, state-dependent diagonal covariance to boost efficiency and fidelity

Target: fewer NFEs while retaining tractable likelihood estimation

Problem Setup: DDPM And DDIM

Consider Markovian DDPM and non-Markovian DDIM samplers

DDPM: reverse transitions Gaussian with learned means and chosen covariances

DDIM: often uses deterministic variance; covariance still crucial

Small T or skip-step sampling sensitive to covariance choices

Aim: analytically optimal, score-informed diagonal covariance estimator

02 Heuristic And Data-Driven Covariances

Modeling Capability

Covariance Type	+#Passes	Intuition
x_t-independent Isotropic β (Ho et al., 2020)	0	Cov. of $q(x_t\|x_{t-1})$
x_t-independent Isotropic $\tilde{\beta}$ (Ho et al., 2020)	0	Cov. of $q(x_t\|x_{t-1}, x_0)$
x_t-independent Isotropic Estimation (Bao et al., 2022b)	0	Estimate from data
x_t-dependent Diagonal VLB (Nichol & Dhariwal, 2021)	1	Learn from data
x_t-dependent Diagonal NS (Bao et al., 2022a)	1	Learn from data
x_t-dependent Diagonal OCM (Ours)	1	Learn from score
x_t-dependent Diagonal Estimation (Zhang et al., 2024b)	$2M$	Estimate from score
x_t-dependent Diagonal Analytic (Zhang et al., 2024b)	D	Calculate from score
x_t-dependent Full Analytic (Zhang et al., 2024b)	D	Calculate from score

- Common options: β/˜β, VLB-learned diagonals, and noise-statistics methods
- Issues: state-independence, error amplification near t→0, ill-defined for some skips
- Struggle to balance FID and likelihood in few-step samplers

02 Analytic Optimal Covariance And Practical Barriers

- Optimal posterior covariance depends on score and Hessian
- Computing diagonal requires O(D)–O(D^2) evals/storage
- Hutchinson estimators reduce storage but need many probes
- Large M makes sampling slow despite gains

03 Key Contributions

Optimal Covariance Matching (OCM) Objective

Unbiased objective regresses Hessian diagonal via Rademacher vectors outside nonlinearity

Upper-bounds ideal loss and is tight at optimum

Accurate diagonal covariance with as few as one probe

Score-Informed, Amortized Diagonal Covariance

Lightweight hϕ predicts Hessian diagonal conditioned on state and time

Inference adds one extra parallel pass with mean/score

Avoids expensive per-step probe averaging

03 General Applicability And Acceleration

- Plug OCM covariances into skip-step DDPM and DDIM
- Improves FID, recall, and likelihood with fewer NFEs
- Works in pixel and latent models, including DiT ImageNet 256×256

04 Method Overview

Generalized Analytical Covariance Identity

For Gaussian convolutions, posterior covariance depends on score and Hessian

Connects to second-order Tweedie identity

Accurate score suffices to infer optimal covariance

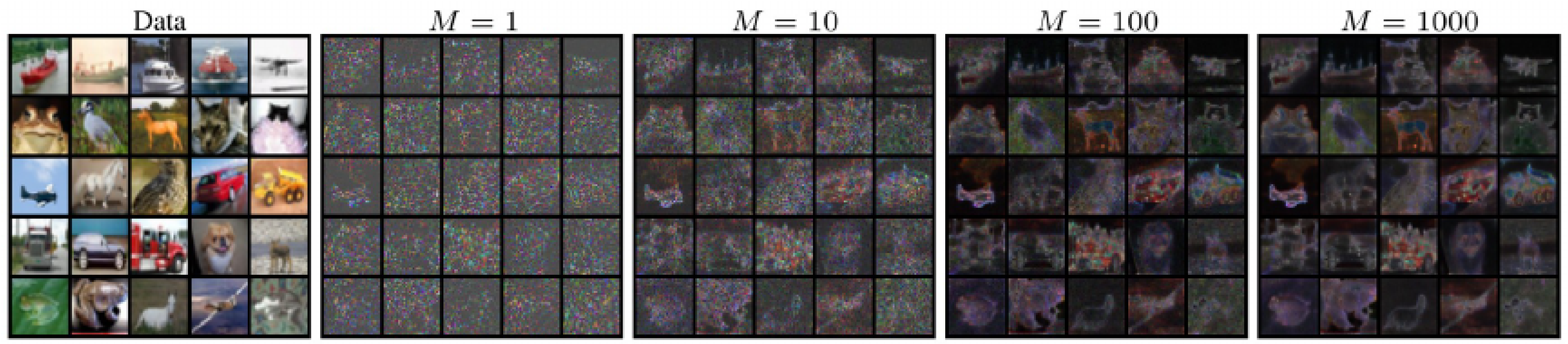
From Hessian Diagonal To Covariance

Approximate Hessian diagonal with hϕ(x,t)

Map to optimal diagonal covariance via analytical identity

Yields state- and time-dependent variances for DDPM/DDIM

05 Unbiased OCM Training Objective

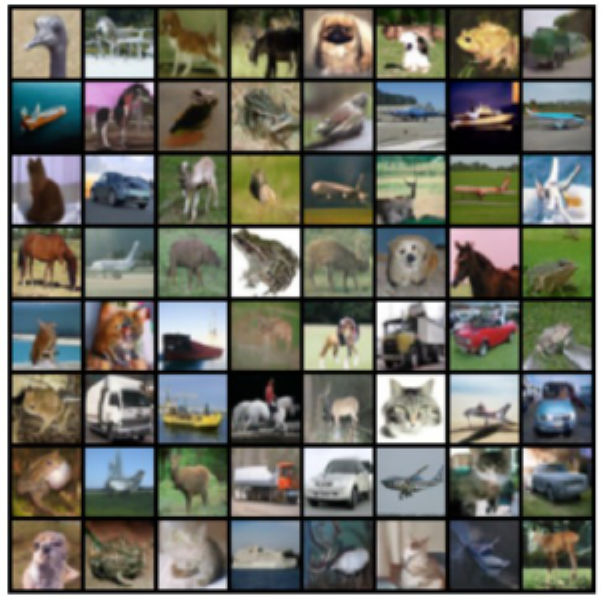


- Moving expectation outside nonlinearity removes bias at small probe counts
- Uses Rademacher vectors to form quadratic forms without full Hessians
- Practically, M=1 suffices for efficient training

05 Network Parameterization And Sharing

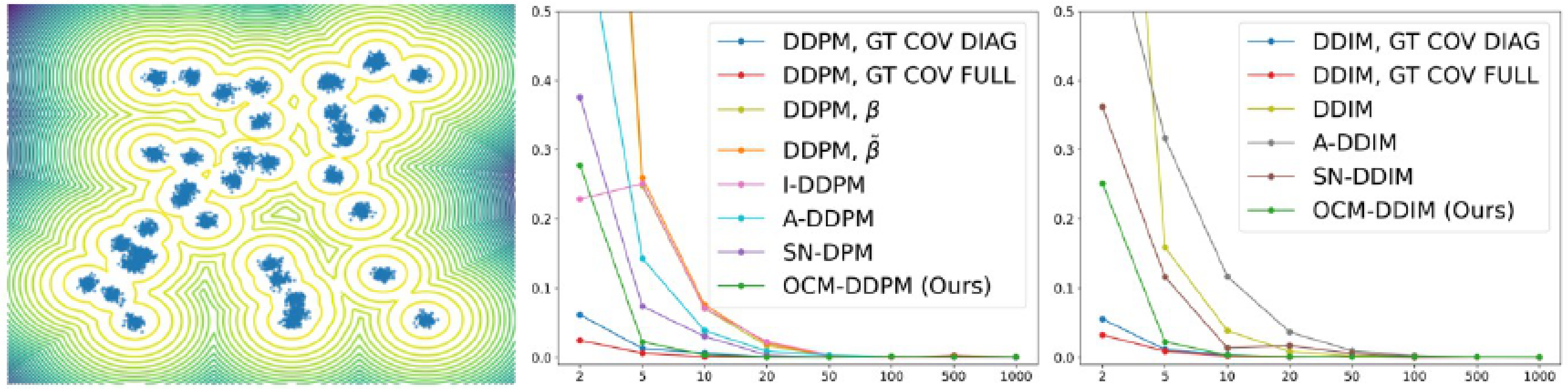
- Reuse pretrained score sθ; add small head hϕ sharing backbone
- Freeze θ and train ϕ via OCM
- Negligible extra compute/memory; joint prediction at inference

05 Skip-Step DDPM And DDIM With OCM



- Closed-form means with pretrained score; covariances from hϕ and identity
- Sampling: compute mean, compute covariance, then sample
- In DDIM, replace delta posterior with Gaussian N(μ0, Σ0)

06 Toy 2D Gaussian Mixture



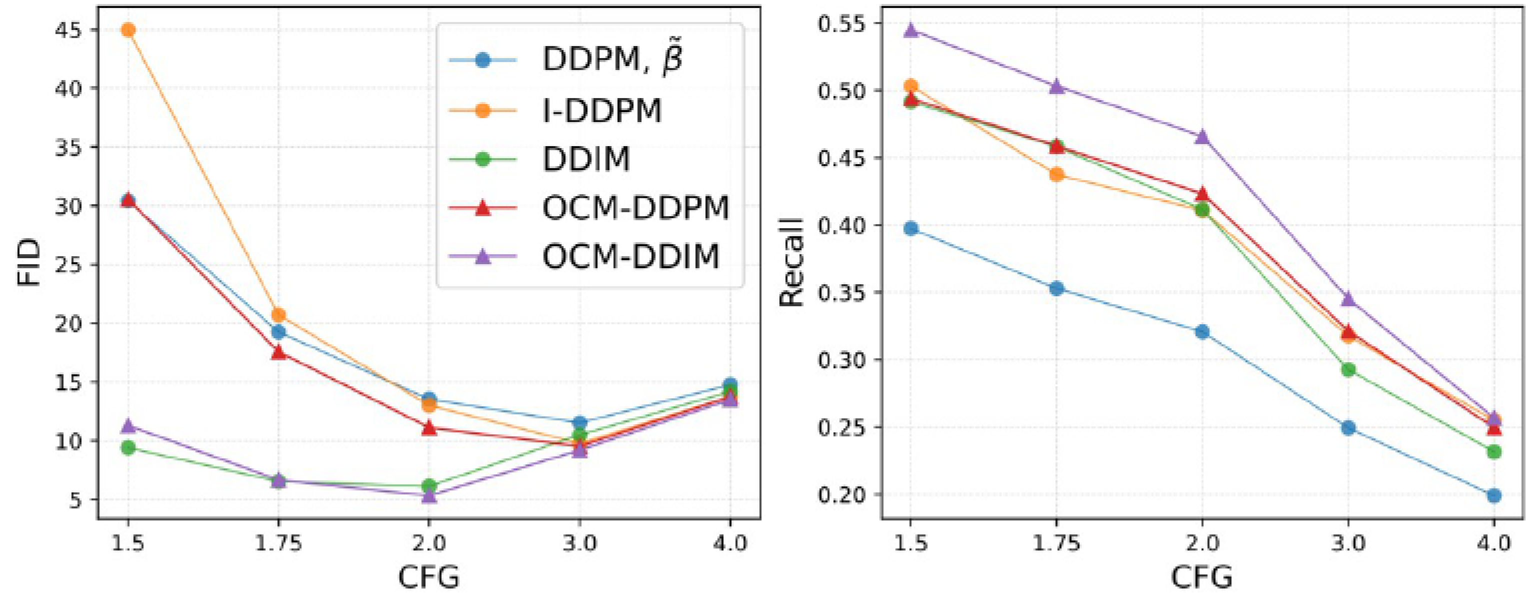
- 2D MoG enables ground-truth scores and covariances for evaluation
- OCM outperforms VLB and noise-statistics diagonals
- Approaches analytic diagonal; full covariance remains upper bound

06 Pixel-Space Image Modeling

	CIFAR10 (LS)						CIFAR10 (CS)					
# TIMESTEPS K	10	25	50	100	200	1000	10	25	50	100	200	1000
DDPM, $\tilde{\beta}$	44.45	21.83	15.21	10.94	8.23	5.11	34.76	16.18	11.11	8.38	6.66	4.92
DDPM, β	233.41	125.05	66.28	31.36	12.96	**3.04**	205.31	84.71	37.35	14.81	5.74	**3.34**
A-DDPM	34.26	11.60	7.25	5.40	4.01	4.03	22.94	8.50	5.50	4.45	4.04	4.31
NPR-DDPM	32.35	10.55	6.18	4.52	3.57	4.10	19.94	7.99	5.31	4.52	4.10	4.27
SN-DDPM	**24.06**	**6.91**	**4.63**	**3.67**	**3.31**	3.65	16.33	6.05	4.17	**3.83**	**3.72**	4.07
OCM-DDPM	24.94	9.19	5.95	4.36	3.48	3.98	**14.32**	**5.54**	**4.10**	3.84	3.75	4.18
DDIM	21.31	10.70	7.74	6.08	5.07	4.13	34.34	16.68	10.48	7.94	6.69	4.89
A-DDIM	14.00	5.81	4.04	3.55	3.39	3.74	26.43	9.96	6.02	4.88	4.92	4.66
NPR-DDIM	13.34	5.38	3.95	3.53	3.42	3.72	22.81	9.47	6.04	5.02	5.06	4.62
SN-DDIM	12.19	**4.28**	**3.39**	**3.23**	**3.22**	**3.65**	17.90	7.36	5.16	4.63	4.63	**4.51**
OCM-DDIM	**10.66**	4.35	3.48	3.27	3.29	3.74	**16.70**	**6.71**	**4.72**	**4.30**	**4.54**	4.53

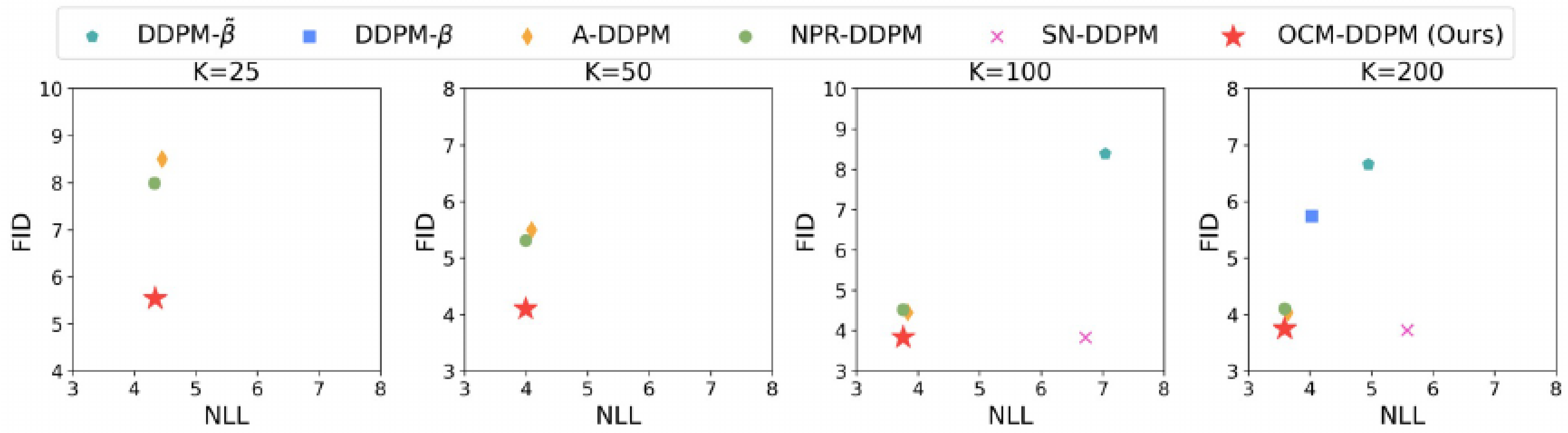
- Datasets: CIFAR10, CelebA 64×64, LSUN Bedroom
- Metrics: FID and ELBO-based NLL across steps
- OCM: competitive/superior FID and best/second-best NLL

06 Latent Diffusion And DiT On ImageNet 256×256



- Replace DiT learned covariance; retain noise-prediction mean
- 10-step sampling with CFG improves FID and Recall
- Fewest steps to reach FID≈6 among covariance methods

07 Likelihood And Sample Quality Trade-Off



- OCM attains strong likelihoods with competitive FID
- SN-DDPM: sometimes better FID, worse likelihood
- NPR-DDPM: better likelihood, lags in FID

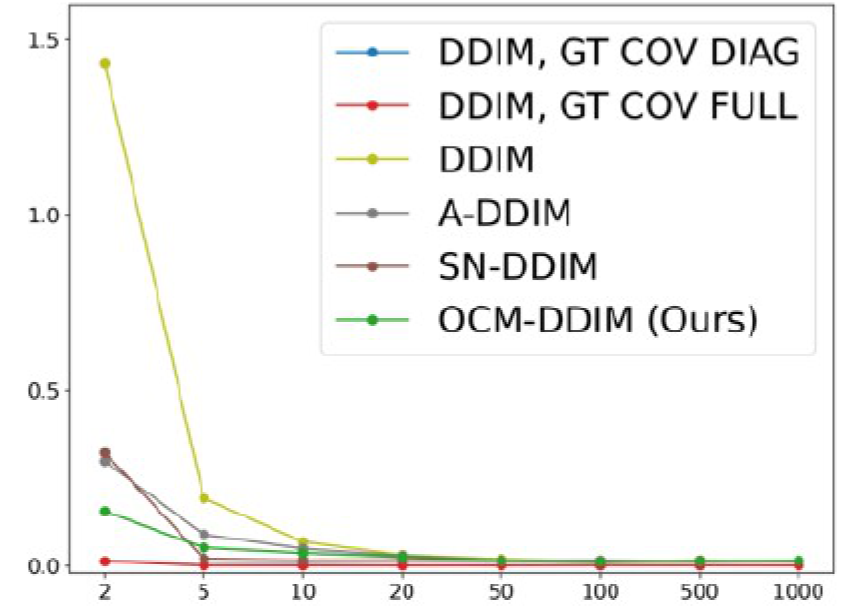
07 Efficiency And Ablations

	FID ↓				NLL (%) ↑			
$K =$	10	25	50	100	10	25	50	100
$M = 1$	14.32	5.54	4.10	3.84	4.99	4.34	3.99	3.76
$M = 3$	14.18	5.51	4.11	3.82	4.99	4.34	3.99	3.76
$M = 5$	14.17	5.51	4.11	3.82	4.99	4.34	3.99	3.76
$M = 10$	14.16	5.51	4.11	3.82	4.99	4.34	3.99	3.76

- Avoids large M at inference; parallel with score pass
- Ablations confirm M=1 suffices during training
- Tables show robust FID/NLL gains across schedules

08 Positioning Among Alternatives

- Heuristics: β/˜β fast but weak; A-DDPM isotropic, data-estimated
- I-DDPM and SN/NPR improve but remain limited
- Score-derived analytic estimators accurate yet expensive
- OCM: score-informed, amortized, supports DDPM/DDIM skips



(d) DDIM MMD v.s. Steps

09 Takeaways And Broader Impact

- OCM: unbiased, efficient route to optimal diagonal covariance
- Improves quality, diversity, and likelihood with fewer steps
- Scales to latent models and large images
- Future: low-rank/block structures, advanced solvers, distillation, and broader applications

THANKS!